

THERAPY-SPEAK IN GENERAL WEB DISCOURSE

ADHD AND AUTISM OVER TIME IN COMMON CRAWL

Jakob Lütke-meier

M.Sc. Applied Social Data Science
Trinity College Dublin

MSc Dissertation Interim Presentation

Research question

How have ADHD- and autism-related terms changed in **prominence**, **meaning**, and **framing** over roughly the last decade?

WHY THIS TOPIC, AND WHY NOW?

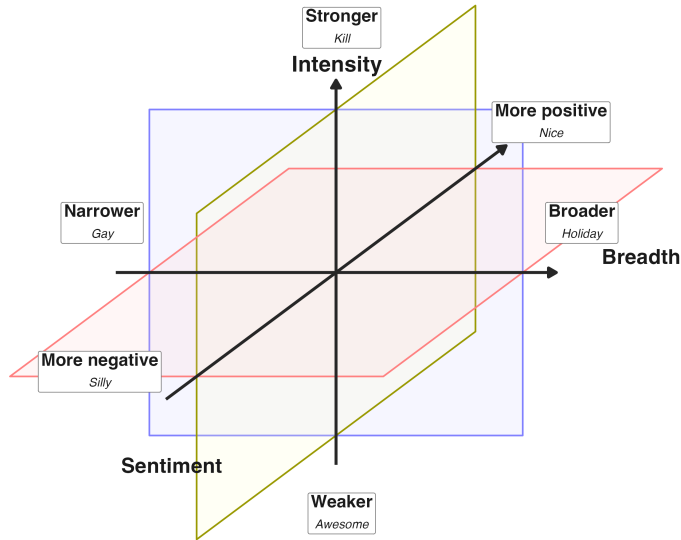
Why this matters

- ▶ Mental-health language is no longer confined to the therapy room
- ▶ That openness is, first and foremost, a good thing
- ▶ But another side of the coin may be **semantic drift**
- ▶ Mis- and overuse can blur meanings and reduce access to the words people actually need
- ▶ ADHD and autism seem especially salient cases

Aim

Move beyond anecdotes toward large-scale evidence

WHAT KINDS OF CHANGE AM I STUDYING?



Other measures of change

Salience

Frequency of use over time

Thematic framing

Topics and contexts of use

HOW THIS BECOMES MEASURABLE

From concept to measure

Concept	Measure	What I look at
Salience	yearly mention rates	how often ADHD/autism terms appear over time
Intensity	collocate arousal / severity	whether surrounding language becomes weaker or stronger
Sentiment	collocate valence	whether surrounding language becomes more positive or negative
Breadth	contextual dispersion	how narrowly or broadly terms are used across contexts
Thematic framing	topic modelling over time	which kinds of topics and associations cluster around the terms

WHY COMMON CRAWL?

10+ PiB

archive size

300B+

web pages

since 2007

historical depth

General web discourse at a scale suitable for diachronic analysis

THE CHALLENGE WITH COMMON CRAWL

Main challenge

Getting **clean text** in a way that is **computationally feasible**

Common Crawl data layers

WET

extracted plaintext only

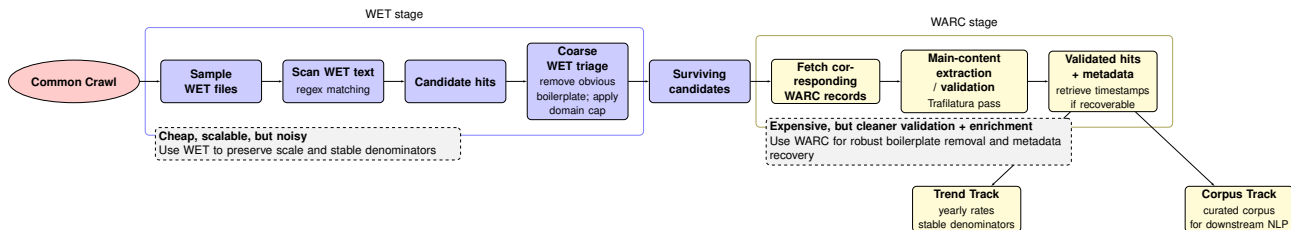
WARC

complete web-page data, including HTML and metadata

Typical boilerplate examples

- ▶ navigation menus
- ▶ cookie banners
- ▶ accessibility widgets

CORPUS COLLECTION PIPELINE



Current position

WET-based screening is working → now moving into WARC-based validation/enrichment

WET stage pilot test results

- ▶ ~157k documents scanned
- ▶ 728 candidate hits (0.46%)
- ▶ 596 surviving candidates post WET triage

NEXT STEPS, RISKS, AND WHAT SUCCESS LOOKS LIKE

Next

- ▶ WARC-based full-fledged boilerplate removal
- ▶ timestamp / metadata enrichment
- ▶ then scale to the full diachronic dataset

Main risks

- ▶ residual boilerplate noise
- ▶ incomplete timestamps
- ▶ computational cost

Success looks like

- ▶ stable yearly trend estimates
- ▶ clean(er) validated corpus